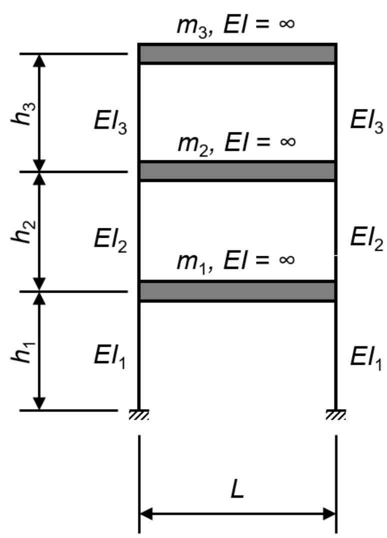
Structural Dynamics Question

Given: Steel building frame shown in the sketch. Based on as-built drawings, the following structural properties have been determined:

- All columns are made of W14x120 sections (assume the columns are oriented to act about their strong axes). Online AISC tables: https://beamdimensions.com/database/American/AISC/W_shapes
- $m_1 = m_2 = 0.24$ kip*s^2/in; $m_3 = 0.12$ kip*s^2/in (= total masses lumped at floor levels)
- Assume a live load-to-dead load ratio = 1:3
- Assume the column masses are accounted for in the lumped masses, m_1 to m_3
- Do not use load factors in your calculations



Tasks (to be performed using MATLAB, Python, or similar coding language):

1. Compute and report the modal properties (= natural vibration frequencies and mode shapes) of the frame. Assume the live load portion varies between 25 and 75% of the specified value. Consider this variability in your computations and how you report the natural vibration frequencies and visualize the mode shapes.
2. Due to exposure to chlorides, the outside flange of the right column in Story 1 has started to corrode. Assume a corrosion rate of 0.25 mm/year and that corroded steel layer can be neglected for structural purposes. You may further assume that corrosion is evenly distributed across the entire height of the column. Compute and report the modal properties of the frame after 20 years of active corrosion.
3. Structural health monitoring (SHM) based on modal properties (= modal analysis) has been suggested by some as a useful tool for damage detection in civil structures. Based on your findings from Tasks 1 and 2, discuss whether such a monitoring approach would be helpful in detecting corrosion progression for this steel frame.

Prepare a short report with your findings; provide calculations and scripts in an appendix.